```
import pandas as pd
import numpy as np
import math
```

```
ds = pd.read_csv("wan_dataset.csv")
```

SIMPLE LINEAR REGRESSION

```
sumx = 0
sumy = 0
num=0
den=0

for x in ds['packet_loss']:
  sumx += x
for y in ds['latency_ms']:
  sumy += y

meanx = sumx / len(ds['packet_loss'])
meany = sumy / len(ds['latency_ms'])

for x,y in zip(ds['packet_loss'],ds['latency_ms']):
  num += (x-meanx)*(y-meany)

for x in ds['packet_loss']:
  den += (x-meanx)**2


w1 = num/den

w0= meany - w1*meanx

print("Linear Regression Coefficient/Slope: ", w1)
print("Linear Regression Intercept: ", w0)
print("Line Equation: y=", w0, "+", w1, "x")
```

```
Linear Regression Coefficient/Slope:  -0.07643613142519337
Linear Regression Intercept:  19.016483844928118
Line Equation: y= 19.016483844928118 + -0.07643613142519337 x
```

MULTIPLE LINEAR REGRESSION

```
x1 = ds['packet_loss'].to_numpy()
x2 = ds['link_utilization'].to_numpy()
x3 = ds['throughput_mbps'].to_numpy()
y = ds['latency_ms'].to_numpy()

X = np.column_stack((np.ones(len(ds)), x1, x2, x3))

XT = X.T

b = np.linalg.lstsq(XT @ X, XT @ y, rcond=None)[0]

b0 = b[0]
b1 = b[1]
b2 = b[2]
b3 = b[3]

print("Multiple Regression Coefficients:")
print("Intercept:", b0)
print("Packet Loss Coefficient:", b1)
print("Link Utilization Coefficient:", b2)
print("Throughput Coefficient:", b3)

print("Regression Equation:")
print("y =", b0, "+", b1, "x1 +", b2, "x2 +", b3, "x3")
```

```
Multiple Regression Coefficients:
Intercept: 10.726496373394323
Packet Loss Coefficient: -0.11880433379892126
Link Utilization Coefficient: 0.07879613971978258
Throughput Coefficient: 0.02148092660217843
```

```
Regression Equation:
y = 10.726496373394323 + -0.11880433379892126 x1 + 0.07879613971978258 x2 + 0.02148092660217843 x3
```